

BERT vs Sentence-BERT: Sample Input/Output

Sample demonstration showing how each model processes the same resume and job description text, and the similarity score each one produces.

Model 1: Sentence-BERT — Used in this project

```
MODEL: Sentence-BERT (all-MiniLM-L6-v2)

INPUT (Resume Text):
"Final year B.Tech AI and Data Science student. Experience
building multi-agent AI pipelines using LangGraph and CrewAI.
Skilled in NLP, embedding models, and sentence similarity search."

INPUT (Job Description):
"Looking for an engineer skilled in machine learning, NLP, and
building intelligent multi-agent systems using modern AI
frameworks."

PROCESSING:
  Step 1: Resume text      -> SBERT encoder -> Vector A (384-dim)
  Step 2: Job description -> SBERT encoder -> Vector B (384-dim)
  Step 3: Cosine Similarity(Vector A, Vector B) computed

OUTPUT:
  Similarity Score          : 0.87    (High Match)
  Spearman Correlation      : 0.721
  Pearson Correlation       : 0.946
  Encoding Time             : 0.044 sec

VERDICT:
  Strong semantic match -- model correctly identifies that the
  resume's AI/NLP skills align with the job description's
  requirements, even though exact words differ ("multi-agent AI
  pipelines" vs "intelligent multi-agent systems").
```

Model 2: Plain BERT — Not used, shown for comparison

MODEL: Plain BERT (bert-base-uncased, mean-pooled)

INPUT (Resume Text):

"Final year B.Tech AI and Data Science student. Experience building multi-agent AI pipelines using LangGraph and CrewAI. Skilled in NLP, embedding models, and sentence similarity search."

INPUT (Job Description):

"Looking for an engineer skilled in machine learning, NLP, and building intelligent multi-agent systems using modern AI frameworks."

PROCESSING:

Step 1: Resume text -> BERT encoder -> Average of all token vectors -> Vector A (768-dim)

Step 2: Job description -> BERT encoder -> Average of all token vectors -> Vector B (768-dim)

Step 3: Cosine Similarity(Vector A, Vector B) computed

OUTPUT:

Similarity Score : 0.69 (Misleading -- close to unrelated text scores too, e.g. Sales role = 0.63)

Spearman Correlation : 0.370

Pearson Correlation : 0.818

Encoding Time : 0.244 sec

VERDICT:

Weak separation -- BERT was never trained to produce comparable sentence vectors, so scores for relevant and irrelevant job roles end up bunched close together, making it unreliable for distinguishing a good match from a bad one.

Side-by-Side Summary

Metric	Plain BERT	Sentence-BERT	Winner
Similarity Score (this example)	0.69	0.87	SBERT
Spearman Correlation	0.370	0.721	SBERT
Pearson Correlation	0.818	0.946	SBERT
Encoding Time	0.244 sec	0.044 sec	SBERT
Separates relevant vs irrelevant roles	No	Yes	SBERT

Conclusion: Sentence-BERT was chosen for TalentSphere AI's resume-to-job matching because it produces sentence embeddings specifically fine-tuned for similarity comparison, giving both higher accuracy and faster performance than plain BERT.